Editor-in-Chief
Physica Scripta

Dear Editor,

I am pleased to submit my manuscript entitled "What do spectral criteria of quantum feature-map kernels actually predict? Classical simulability, generalization advantage, and the role of task difficulty" for consideration for publication as a regular article in Physica Scripta.

The manuscript reports a task-resolved numerical study (721 configurations, exact state-vector simulation) that separates two questions often conflated in quantum machine learning: whether spectral criteria of quantum feature-map kernels predict classical simulability (they do, reliably), versus whether they predict generalization advantage (they do not; the sign is unstable across tasks). The dominant predictor of advantage is the saturation of the classical baselines, which we quantify. The study includes a counterexample (amplitude damping) and a null result for kernel–target alignment, and argues for concrete evaluation guidelines for quantum kernel studies.

The work is single-authored, has not been published previously in the peer-reviewed literature, and is not under consideration for publication elsewhere. The complete code and all data are openly available at https://github.com/python123flask/quantum-kernel-spectral. I have no conflicts of interest to declare, and no external funding was received for this work.

Thank you for considering this submission. I would be glad to provide any further information upon request.

Sincerely,

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